

BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC Mathematics Comprehensive Study Plan

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Real Analysis: MCQs with Full Explanations

Topics Covered:

- Sets, Fields & Ordered Fields
- Sequences & Their Limits
- Cauchy Sequences
- Tests of Convergence
- Continuity & Uniform Continuity
- Mean Value Theorems
- Riemann Integration
- Sequences & Series of Functions
- The Real Number System
- Monotone Sequences
- Series of Real Numbers
- Limits of Functions
- Differentiability
- Taylor's Theorem
- Improper Integrals
- Uniform Convergence

Each section: **Concept Summary** → **Key Definitions & Formulas**
→ **MCQs with Full Explanations**

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1. The Real Number System

The real number system $\mathbb{R}$ is a **complete ordered field**. Key properties include:

- **Completeness (Least Upper Bound Property):** Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound (supremum) in $\mathbb{R}$.
- **Archimedean Property:** For any $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.
- **Density of $\mathbb{Q}$:** Between any two real numbers, there exists a rational number.

Key Definitions & Results

- $\sup A$ = least upper bound; $\inf A$ = greatest lower bound
- If $A \subseteq B$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$
- $\sup(-A) = -\inf(A)$ and $\inf(-A) = -\sup(A)$
- $|x| < \varepsilon \iff -\varepsilon < x < \varepsilon$
- Triangle inequality: $|x + y| \leq |x| + |y|$
- Reverse triangle inequality: $||x| - |y|| \leq |x - y|$

1.1. MCQs on Real Number System

Q1. The least upper bound of the set $A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$ is:

- (A) 0
- (B) $\frac{1}{2}$
- (C) 1
- (D) Does not exist

Correct Answer: (C) 1

The set $A = \left\{ \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots \right\}$. Each term $a_n = \frac{n}{n+1} = 1 - \frac{1}{n+1} < 1$ for all n , so 1 is an upper bound. For any $M < 1$, we have $1 - M > 0$, and by the Archimedean property there exists n such that $\frac{1}{n+1} < 1 - M$, i.e., $a_n > M$. Thus 1 is the least upper bound.

Q2. Which of the following properties is equivalent to the completeness of $\mathbb{R}$?

- (A) Every bounded sequence converges
- (B) Every Cauchy sequence converges in $\mathbb{R}$
- (C) Every non-empty bounded set has a supremum in $\mathbb{R}$
- (D) All of the above

Correct Answer: (D) All of the above

All three properties — (i) Bolzano-Weierstrass (every bounded sequence has a convergent subsequence), (ii) every Cauchy sequence converges, and (iii) the Least Upper Bound Property — are equivalent characterizations of the completeness of $\mathbb{R}$. Each can be used to prove the others.

Q3. For any two real numbers $a < b$, which statement is TRUE?

- (A) There is no irrational number between a and b
- (B) There exists a rational number q with $a < q < b$
- (C) $b - a$ must be rational
- (D) There are only finitely many rationals between a and b

Correct Answer: (B)

This is the **Density of $\mathbb{Q}$ in $\mathbb{R}$** : between any two distinct real numbers, there exists at least one rational number. By the Archimedean property, choose $n \in \mathbb{N}$ with $n(b - a) > 1$, then pick the integer m with $na < m < nb$, giving $q = m/n \in (a, b)$.

Q4. If $A \subseteq \mathbb{R}$ is bounded above and $B = \{-a : a \in A\}$, then:

- (A) $\sup B = \sup A$
- (B) $\sup B = -\sup A$
- (C) $\inf B = -\inf A$
- (D) $\sup B = -\inf A$

Correct Answer: (D) $\sup B = -\inf A$

Since $B = -A$, we have $\sup(-A) = -\inf(A)$. For example, if $A = [-3, 5]$, then $B = [-5, 3]$, $\inf A = -3$, and $\sup B = 3 = -(-3) = -\inf A$. ✓

Remember the reverse triangle inequality: $||x| - |y|| \leq |x - y|$. This is used frequently in proofs involving limits and continuity in PSC exams.

Q5. The Archimedean property states that for any $x \in \mathbb{R}$:

- (A) $\exists n \in \mathbb{N}$ such that $n > x$
- (B) $\exists n \in \mathbb{N}$ such that $\frac{1}{n} > x$ for all $x > 0$
- (C) For any $x > 0$, $\exists n \in \mathbb{N}$ with $\frac{1}{n} < x$
- (D) Both (A) and (C)

Correct Answer: (D) Both (A) and (C)

The standard Archimedean property says: $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}$ with $n > x$. An equivalent form: $\forall x > 0, \exists n \in \mathbb{N}$ with $\frac{1}{n} < x$ (choose $n > 1/x$). Both are valid and equivalent statements of the Archimedean property.

2. Sequences of Real Numbers

A sequence (a_n) **converges** to $L \in \mathbb{R}$ if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } n > N \implies |a_n - L| < \varepsilon$$

Key types:

- **Monotone increasing:** $a_n \leq a_{n+1}$ for all n
- **Bounded:** $\exists M > 0$ with $|a_n| \leq M$ for all n
- **Cauchy:** $\forall \varepsilon > 0, \exists N$ such that $m, n > N \implies |a_m - a_n| < \varepsilon$

Key Definitions & Results

- **Monotone Convergence Theorem:** A monotone bounded sequence converges.
- **Bolzano–Weierstrass Theorem:** Every bounded sequence has a convergent subsequence.
- **Squeeze Theorem:** $a_n \leq b_n \leq c_n$ and $a_n, c_n \rightarrow L \implies b_n \rightarrow L$.
- $\lim_{n \rightarrow \infty} r^n = 0$ if $|r| < 1$; diverges if $|r| > 1$.
- $\lim_{n \rightarrow \infty} n^{1/n} = 1$; $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$.
- $\lim_{n \rightarrow \infty} \frac{n!}{n^n} = 0$; $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$ for any fixed x .

2.1. MCQs on Sequences

Q6. The sequence $a_n = \frac{(-1)^n}{n}$ is:

- (A) Monotone and convergent
- (B) Convergent but not monotone
- (C) Divergent and bounded
- (D) Monotone and bounded but divergent

Correct Answer: (B) Convergent but not monotone

We have $a_n = \frac{(-1)^n}{n}$. Since $|a_n| = \frac{1}{n} \rightarrow 0$, the sequence converges to 0 by the Squeeze Theorem: $-\frac{1}{n} \leq a_n \leq \frac{1}{n} \rightarrow 0$. It is not monotone because $a_1 = -1$, $a_2 = \frac{1}{2}$, $a_3 = -\frac{1}{3}$, alternating in sign.

Q7. If (a_n) is a Cauchy sequence of real numbers, then it is:

- (A) Monotone
- (B) Bounded and convergent
- (C) Convergent but not necessarily bounded
- (D) Bounded but not necessarily convergent

Correct Answer: (B) Bounded and convergent

In $\mathbb{R}$: (i) Every Cauchy sequence is bounded (take $\varepsilon = 1$, find N , then the sequence is bounded by $\max\{|a_1|, \dots, |a_N|, |a_N| + 1\}$). (ii) By the completeness of $\mathbb{R}$, every Cauchy sequence converges. Note: In general metric spaces, Cauchy sequences may not converge; $\mathbb{R}$ is *complete* precisely because they do.

Q8. $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$ equals:

- (A) 1
- (B) x
- (C) e^x
- (D) xe

Correct Answer: (C) e^x

This is the standard limit: $\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$ for all $x \in \mathbb{R}$. For $x = 1$ it gives the definition of $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718 \dots$

Q9. The Bolzano–Weierstrass Theorem states that:

- (A) Every sequence in $\mathbb{R}$ is convergent
- (B) Every bounded sequence in $\mathbb{R}$ has a convergent subsequence
- (C) Every Cauchy sequence in $\mathbb{R}$ converges
- (D) Every monotone sequence in $\mathbb{R}$ converges

Correct Answer: (B)

The Bolzano–Weierstrass Theorem: *Every bounded sequence in $\mathbb{R}$ has at least one convergent subsequence.* This is a fundamental theorem and a consequence of completeness. Note that the sequence itself need not converge (e.g., $a_n = (-1)^n$ is bounded, but diverges; its subsequences $a_{2n} = 1$ and $a_{2n-1} = -1$ both converge).

Q10. If $\lim_{n \rightarrow \infty} a_n = L$ and $\lim_{n \rightarrow \infty} b_n = M$, then $\lim_{n \rightarrow \infty} (a_n b_n)$ is:

- (A) $L + M$
- (B) LM

- (C) L/M (provided $M \neq 0$)
 (D) It may not exist

Correct Answer: (B) LM

By the **Algebra of Limits**: if $a_n \rightarrow L$ and $b_n \rightarrow M$, then $a_n + b_n \rightarrow L + M$, $a_nb_n \rightarrow LM$, and $a_n/b_n \rightarrow L/M$ (when $M \neq 0$). These follow from the ε - N definition of limits.

To prove a sequence does NOT converge, find two subsequences converging to different limits. For example, $a_n = (-1)^n$ diverges because $a_{2n} \rightarrow 1$ and $a_{2n-1} \rightarrow -1$.

Q11. A sequence (a_n) is monotone increasing and bounded above. Then:

- (A) It must be Cauchy but not necessarily convergent
 (B) It converges to $\sup\{a_n : n \in \mathbb{N}\}$
 (C) It converges to $\inf\{a_n : n \in \mathbb{N}\}$
 (D) It diverges to $+\infty$

Correct Answer: (B)

By the **Monotone Convergence Theorem**: a monotone increasing sequence bounded above converges, and its limit equals the supremum of its range, i.e., $\lim_{n \rightarrow \infty} a_n = \sup\{a_n : n \in \mathbb{N}\}$. Similarly, a monotone decreasing sequence bounded below converges to the infimum.

3. Series of Real Numbers

A series $\sum_{n=1}^{\infty} a_n$ **converges** if the sequence of partial sums $S_n = \sum_{k=1}^n a_k$ converges. Otherwise it diverges.

Necessary condition: If $\sum a_n$ converges, then $a_n \rightarrow 0$. The converse is *FALSE* (e.g., harmonic series $\sum \frac{1}{n}$ diverges even though $\frac{1}{n} \rightarrow 0$).

Key Definitions & Results

- **Geometric Series:** $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$ if $|r| < 1$; diverges if $|r| \geq 1$.
- **p -Series:** $\sum \frac{1}{n^p}$ converges $\iff p > 1$.

- **Ratio Test:** $L = \lim \left| \frac{a_{n+1}}{a_n} \right|$: converges if $L < 1$; diverges if $L > 1$; inconclusive if $L = 1$.
- **Root Test:** $L = \lim |a_n|^{1/n}$: converges if $L < 1$; diverges if $L > 1$.
- **Comparison Test:** $0 \leq a_n \leq b_n$: $\sum b_n$ converges $\implies \sum a_n$ converges.
- **Alternating Series Test (Leibniz):** $\sum (-1)^n b_n$ converges if $b_n \searrow 0$.
- **Integral Test:** $\sum f(n)$ and $\int_1^\infty f(x) dx$ converge or diverge together.

3.1. MCQs on Series

Q12. The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is:

- (A) Divergent
- (B) Convergent, sum $= \pi^2/6$
- (C) Convergent, sum $= 1$
- (D) Convergent, sum $= \pi/4$

Correct Answer: (B) Convergent, sum $= \pi^2/6$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a p -series with $p = 2 > 1$, so it converges. Euler famously proved the **Basel Problem**: $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. This should be memorized for exams.

Q13. The series $\sum_{n=1}^{\infty} \frac{1}{n}$ (harmonic series):

- (A) Converges to $\ln 2$
- (B) Converges to 1
- (C) Diverges to $+\infty$
- (D) Oscillates and diverges

Correct Answer: (C) Diverges to $+\infty$

Even though $\frac{1}{n} \rightarrow 0$, the harmonic series diverges. Proof by grouping (Cauchy's condensation): $1 + \frac{1}{2} + (\frac{1}{3} + \frac{1}{4}) + (\frac{1}{5} + \dots + \frac{1}{8}) + \dots > 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \rightarrow \infty$.

Q14. By the Ratio Test, the series $\sum_{n=0}^{\infty} \frac{n!}{2^n}$:

- (A) Converges
- (B) Diverges
- (C) Test is inconclusive

(D) Converges to e^2

Correct Answer: (B) Diverges

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)!}{2^{n+1}} \cdot \frac{2^n}{n!} = \lim_{n \rightarrow \infty} \frac{n+1}{2} = +\infty > 1.$$

Since $L > 1$, the series diverges by the Ratio Test.

Q15. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$:

- (A) Diverges
- (B) Converges absolutely
- (C) Converges conditionally to $\ln 2$
- (D) Converges conditionally to $\pi/4$

Correct Answer: (C) Converges conditionally to $\ln 2$

By the **Alternating Series Test**: $b_n = \frac{1}{n}$ is decreasing and $b_n \rightarrow 0$, so the series converges. However, $\sum \frac{1}{n}$ diverges, so convergence is only *conditional*. The sum equals $\ln 2 \approx 0.693$, which follows from the Taylor series of $\ln(1+x)$ at $x = 1$.

Q16. The Cauchy condensation test states that if $a_n \geq 0$ is decreasing, then $\sum a_n$ converges iff $\sum 2^n a_{2^n}$ converges. Using this, $\sum \frac{1}{n(\ln n)^p}$ (for $n \geq 2$) converges iff:

- (A) $p > 1$
- (B) $p \geq 1$
- (C) $p > 0$
- (D) $p > 2$

Correct Answer: (A) $p > 1$

Apply Cauchy condensation: $\sum 2^n \cdot \frac{1}{2^n(\ln 2^n)^p} = \sum \frac{1}{(n \ln 2)^p} = \frac{1}{(\ln 2)^p} \sum \frac{1}{n^p}$. This is a p -series, which converges iff $p > 1$.

4. Limits and Continuity

$f : A \rightarrow \mathbb{R}$ is **continuous** at $c \in A$ if:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon$$

f is **uniformly continuous** on A if the same δ works for all $c \in A$ simultaneously.

Key Definitions & Results

- **Extreme Value Theorem:** f continuous on $[a, b]$ attains its max and min.
- **Intermediate Value Theorem (IVT):** If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then $\exists c \in (a, b)$ with $f(c) = 0$.
- **Uniform Continuity:** f is uniformly continuous on $[a, b]$ if f is continuous on $[a, b]$.
- Lipschitz $\implies$ Uniformly continuous $\implies$ Continuous.
- $f(x) = \sin x$ and $f(x) = \cos x$ are uniformly continuous on $\mathbb{R}$.
- $f(x) = x^2$ is continuous but NOT uniformly continuous on $\mathbb{R}$.

4.1. MCQs on Continuity

Q17. The function $f(x) = \frac{\sin x}{x}$ for $x \neq 0$ can be made continuous at $x = 0$ by setting $f(0) =$:

- (A) 0
- (B) π
- (C) 1
- (D) ∞

Correct Answer: (C) 1

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (a fundamental limit). Setting $f(0) = 1$ makes f continuous at 0. This limit can be proved using the squeeze theorem: $\cos x \leq \frac{\sin x}{x} \leq 1$ for $0 < x < \pi/2$, and $\cos x \rightarrow 1$.

Q18. Which of the following functions is uniformly continuous on $\mathbb{R}$?

- (A) $f(x) = x^2$
- (B) $f(x) = e^{x^2}$
- (C) $f(x) = \sin x$
- (D) $f(x) = \tan x$

Correct Answer: (C) $f(x) = \sin x$

$f(x) = \sin x$ is uniformly continuous on $\mathbb{R}$ since $|\sin x - \sin y| \leq |x - y|$ (Lipschitz with constant 1). $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$ because its derivative is unbounded. $\tan x$ is not even continuous on all of $\mathbb{R}$.

Q19. By the Intermediate Value Theorem, if f is continuous on $[0, 1]$ with $f(0) = -2$ and $f(1) = 3$, then:

- (A) f has a maximum at some $c \in (0, 1)$
- (B) $\exists c \in (0, 1)$ such that $f(c) = 0$

- (C) f is differentiable on $(0, 1)$
 (D) f is monotone on $[0, 1]$

Correct Answer: (B)

By the **IVT**: since $f(0) = -2 < 0 < 3 = f(1)$, and f is continuous on $[0, 1]$, there must exist at least one $c \in (0, 1)$ with $f(c) = 0$. This guarantees a root exists, but says nothing about differentiability or monotonicity.

Q20. If f is continuous on a closed bounded interval $[a, b]$, then:

- (A) f must be differentiable on (a, b)
 (B) f must be monotone
 (C) f attains its maximum and minimum values on $[a, b]$
 (D) f is uniformly continuous only if f' is bounded

Correct Answer: (C)

This is the **Extreme Value Theorem**: if f is continuous on a closed, bounded interval $[a, b]$, then f is bounded and attains both its supremum and infimum on $[a, b]$. Also, by the Heine-Cantor theorem, f is automatically uniformly continuous on $[a, b]$.

Key distinction: Uniform continuity requires ONE δ for ALL points, whereas ordinary continuity allows a different δ at each point. Always ask: is the domain a closed bounded interval? If yes, continuity implies uniform continuity.

5. Differentiability

f is **differentiable at** c if $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ exists.

Differentiable $\implies$ Continuous, but Continuous $\not\implies$ Differentiable (e.g., $f(x) = |x|$ at $x = 0$).

Key Definitions & Results

- **Rolle's Theorem**: f continuous on $[a, b]$, differentiable on (a, b) , $f(a) = f(b) \implies \exists c \in (a, b)$ with $f'(c) = 0$.
- **Mean Value Theorem (MVT)**: $\exists c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

- **Cauchy MVT:** $\exists c$ s.t. $\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$.
- **L'Hôpital's Rule:** If $\frac{f(x)}{g(x)} \rightarrow \frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$ (when the latter exists).
- **Taylor's Theorem:** $f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k + R_n(x)$, where $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - a)^{n+1}$.

5.1. MCQs on Differentiability

Q21. The function $f(x) = |x|$ at $x = 0$ is:

- (A) Differentiable and continuous
- (B) Continuous but not differentiable
- (C) Neither continuous nor differentiable
- (D) Differentiable but not continuous

Correct Answer: (B) Continuous but not differentiable

$\lim_{x \rightarrow 0} |x| = 0 = f(0)$, so f is continuous at 0. However, the left derivative is $\lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1$ and the right derivative is $\lim_{h \rightarrow 0^+} \frac{h}{h} = 1$. Since left $\neq$ right, f is not differentiable at 0.

Q22. By the Mean Value Theorem applied to $f(x) = x^3$ on $[0, 3]$, the value of c is:

- (A) 1
- (B) $\sqrt{3}$
- (C) 2
- (D) 3

Correct Answer: (B) $\sqrt{3}$

$f'(c) = \frac{f(3) - f(0)}{3 - 0} = \frac{27 - 0}{3} = 9$. Since $f'(x) = 3x^2$, we solve $3c^2 = 9 \implies c^2 = 3 \implies c = \sqrt{3} \in (0, 3)$. ✓

Q23. Rolle's Theorem requires which of the following conditions?

- (A) f is continuous on $[a, b]$ only
- (B) f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$
- (C) f is differentiable on $[a, b]$ and $f'(a) = f'(b)$
- (D) f is continuous and strictly monotone on $[a, b]$

Correct Answer: (B)

Rolle's Theorem requires three conditions: (1) f continuous on the *closed* interval $[a, b]$, (2) f differentiable on the *open* interval (a, b) , and (3) $f(a) = f(b)$. Under these hypotheses, $\exists c \in (a, b)$ with $f'(c) = 0$.

Q24. Using L'Hôpital's Rule, $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} =$

- (A) 0
- (B) 1
- (C) 1/2
- (D) ∞

Correct Answer: (C) 1/2

Form $\frac{0}{0}$. Apply L'Hôpital once: $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$. Still $\frac{0}{0}$. Apply again: $\lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$. Alternatively, from the Taylor expansion: $e^x = 1 + x + \frac{x^2}{2!} + \dots$, so $\frac{e^x - 1 - x}{x^2} \rightarrow \frac{x^2/2}{x^2} = \frac{1}{2}$.

Q25. The Taylor series of $f(x) = \cos x$ about $x = 0$ is:

- (A) $1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$
- (B) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$
- (C) $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$
- (D) $1 + x + \frac{x^2}{2!} + \dots$

Correct Answer: (B)

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

Note: (C) is $\sin x$, (D) is e^x . Remember that $\cos x$ has only even powers due to being an even function.

6. Riemann Integration

A bounded function f on $[a, b]$ is **Riemann integrable** if $\inf\{U(f, P)\} = \sup\{L(f, P)\}$ (upper and lower Darboux sums agree). Key result: f is Riemann integrable iff it is **bounded and continuous almost everywhere** (Lebesgue's criterion).

Key Definitions & Results

- Every continuous function on $[a, b]$ is Riemann integrable.
- Every monotone bounded function on $[a, b]$ is Riemann integrable.
- **Fundamental Theorem of Calculus (FTC I):** If $F'(x) = f(x)$, then $\int_a^b f(x) dx = F(b) - F(a)$.
- **FTC II:** If f is continuous on $[a, b]$, then $G(x) = \int_a^x f(t) dt$ is differentiable and $G'(x) = f(x)$.
- **Improper Integral:** $\int_1^\infty \frac{1}{x^p} dx$ converges iff $p > 1$; $\int_0^1 \frac{1}{x^p} dx$ converges iff $p < 1$.

6.1. MCQs on Riemann Integration

Q26. Which of the following functions is NOT Riemann integrable on $[0, 1]$?

- (A) $f(x) = x^2$
- (B) $f(x) = |x - \frac{1}{2}|$
- (C) $f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \notin \mathbb{Q} \end{cases}$ (Dirichlet function)
- (D) $f(x) = \sin(\frac{1}{x})$ for $x > 0$, $f(0) = 0$

Correct Answer: (C) Dirichlet function

The Dirichlet function is discontinuous *everywhere* on $[0, 1]$, so $U(f, P) = 1$ and $L(f, P) = 0$ for every partition P . Thus it is not Riemann integrable. Functions (A), (B), and (D) are Riemann integrable (they have at most countably many discontinuities).

Q27. By the Fundamental Theorem of Calculus, if $G(x) = \int_0^x \cos(t^2) dt$, then $G'(x) =$:

- (A) $\sin(x^2)$
- (B) $\cos(x^2)$
- (C) $-\sin(x^2)$
- (D) $2x \cos(x^2)$

Correct Answer: (B) $\cos(x^2)$

By FTC Part II: if $G(x) = \int_0^x f(t) dt$ and f is continuous, then $G'(x) = f(x)$. Here $f(t) = \cos(t^2)$ is continuous, so $G'(x) = \cos(x^2)$.

Q28. The improper integral $\int_1^\infty \frac{1}{x^{3/2}} dx$:

- (A) Diverges
- (B) Converges to 2
- (C) Converges to $\frac{1}{2}$
- (D) Converges to 1

Correct Answer: (B) Converges to 2

$$\int_1^{\infty} x^{-3/2} dx = \left[\frac{x^{-1/2}}{-1/2} \right]_1^{\infty} = -2 \left[\frac{1}{\sqrt{x}} \right]_1^{\infty} = -2(0 - 1) = 2.$$

Since $p = 3/2 > 1$, convergence is expected.

Q29. If f and g are Riemann integrable on $[a, b]$, which of the following is NOT necessarily Riemann integrable?

- (A) $f + g$
- (B) $f \cdot g$
- (C) $|f|$
- (D) f/g

Correct Answer: (D) f/g

$f + g$, $f \cdot g$, $|f|$, and cf are all Riemann integrable whenever f and g are. However, f/g may fail to be bounded (if g is zero somewhere on $[a, b]$), and an unbounded function cannot be Riemann integrable in the classical sense.

7. Sequences and Series of Functions

A sequence of functions (f_n) **converges uniformly** to f on A if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ (independent of } x) \text{ s.t. } n > N \implies |f_n(x) - f(x)| < \varepsilon \text{ for all } x \in A.$$

Pointwise convergence allows N to depend on x ; uniform convergence does not.

Key Definitions & Results

- **Uniform convergence $\implies$** : Limit of continuous functions is continuous; interchange of limit and integral; interchange of limit and derivative (with extra hypothesis).
- **Weierstrass M-Test:** $|f_n(x)| \leq M_n$ for all x and $\sum M_n < \infty \implies \sum f_n$ converges uniformly and absolutely.

- $f_n(x) = x^n$ on $[0, 1]$: converges pointwise to $\begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \end{cases}$, but NOT uniformly.
- Power series $\sum a_n x^n$ converges uniformly on $[-r, r]$ for any $r < R$ (radius of convergence).

7.1. MCQs on Uniform Convergence

Q30. The sequence $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to f where:

- (A) $f(x) = 0$ for all $x \in [0, 1]$
- (B) $f(x) = 1$ for all $x \in [0, 1]$
- (C) $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$
- (D) $f(x)$ does not exist

Correct Answer: (C)

For $0 \leq x < 1$: $\lim_{n \rightarrow \infty} x^n = 0$. For $x = 1$: $\lim_{n \rightarrow \infty} 1^n = 1$. So the pointwise limit is $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. The convergence is NOT uniform because each f_n is continuous but f is discontinuous at $x = 1$ (and uniform convergence preserves continuity).

Q31. The Weierstrass M -Test is used to prove:

- (A) Pointwise convergence of a sequence of functions
- (B) Uniform convergence of a series of functions
- (C) Absolute convergence of a numerical series
- (D) Convergence of improper integrals

Correct Answer: (B) Uniform convergence of a series of functions

The M -Test: if $|f_n(x)| \leq M_n$ for all $x \in A$ and $\sum_{n=1}^{\infty} M_n < \infty$, then $\sum_{n=1}^{\infty} f_n(x)$ converges both *absolutely* and *uniformly* on A . It is one of the most useful tools for establishing uniform convergence of series.

Q32. If (f_n) converges uniformly to f on $[a, b]$ and each f_n is continuous, then:

- (A) f may be discontinuous at some points
- (B) $\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b f(x) dx$
- (C) f must be differentiable on (a, b)
- (D) (f_n) must also be monotone

Correct Answer: (B)

Uniform convergence preserves two key properties: (i) **Continuity** — the limit f is continuous. (ii) **Integration** — limits and integrals can be interchanged: $\lim_{n \rightarrow \infty} \int_a^b f_n = \int_a^b \lim_{n \rightarrow \infty} f_n = \int_a^b f$. This fails for mere pointwise convergence.

Q33. The power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has radius of convergence:

- (A) $R = 1$
- (B) $R = e$
- (C) $R = 0$
- (D) $R = \infty$

Correct Answer: (D) $R = \infty$

By the Ratio Test: $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}/(n+1)!}{x^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1$ for all $x \in \mathbb{R}$. Therefore the series converges for all x , giving $R = \infty$. This is the power series for e^x .

Uniform vs. Pointwise Convergence: To check if $f_n \rightarrow f$ uniformly, compute $\sup_{x \in A} |f_n(x) - f(x)|$ and check if it $\rightarrow 0$ as $n \rightarrow \infty$. If this sup does NOT go to zero, convergence is only pointwise.

8. Additional Mixed MCQs

8.1. Assorted Real Analysis MCQs

Q34. If $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $|f(x) - f(y)| \leq |x - y|^2$ for all $x, y \in \mathbb{R}$, then f is:

- (A) Continuous but not differentiable
- (B) Uniformly continuous but not differentiable
- (C) Differentiable and $f'(x) = 0$ everywhere
- (D) Nowhere differentiable

Correct Answer: (C) Differentiable and $f'(x) = 0$ everywhere

At any x : $\left| \frac{f(x+h) - f(x)}{h} \right| \leq \frac{|h|^2}{|h|} = |h| \rightarrow 0$. So $f'(x) = 0$ for all x . A function with zero derivative everywhere is constant. Also, $|x - y|^2$ type conditions (Hölder with exponent > 1) force f to be constant.

Q35. A function f is said to be of **bounded variation** on $[a, b]$ if:

- (A) f is bounded on $[a, b]$
- (B) $\sup_P \sum_{i=1}^n |f(x_i) - f(x_{i-1})| < \infty$
- (C) f is differentiable on (a, b)
- (D) $\int_a^b |f(x)| dx < \infty$

Correct Answer: (B)

f is of bounded variation on $[a, b]$ if the total variation $V(f; a, b) = \sup_P \sum_{i=1}^n |f(x_i) - f(x_{i-1})| < \infty$, where the supremum is over all partitions P of $[a, b]$. Every monotone function and every Lipschitz function is of bounded variation.

Q36. $\lim_{n \rightarrow \infty} n^{1/n} =$

- (A) e
- (B) 0
- (C) 1
- (D) ∞

Correct Answer: (C) 1

Let $a_n = n^{1/n}$. Take $\ln a_n = \frac{\ln n}{n} \rightarrow 0$ (by L'Hôpital or squeeze). Therefore $a_n = e^{\ln a_n} \rightarrow e^0 = 1$. This is a standard limit that appears frequently in PSC exams.

Q37. If $\sum a_n$ converges absolutely, then:

- (A) a_n is eventually monotone
- (B) $\sum a_n$ converges for any rearrangement
- (C) a_n must be positive
- (D) $\sum a_n^2$ may diverge

Correct Answer: (B)

By the **Riemann Rearrangement Theorem**: if $\sum a_n$ converges *absolutely*, then every rearrangement also converges to the same sum. If $\sum a_n$ converges only conditionally, rearrangements can converge to any value or even diverge.

Q38. The set of rational numbers $\mathbb{Q}$ is:

- (A) Closed in $\mathbb{R}$
- (B) Open in $\mathbb{R}$
- (C) Neither open nor closed in $\mathbb{R}$

(D) Both open and closed in $\mathbb{R}$

Correct Answer: (C) Neither open nor closed in $\mathbb{R}$

$\mathbb{Q}$ is *not open*: any open interval (a, b) contains irrationals, so no open set can be contained in $\mathbb{Q}$. $\mathbb{Q}$ is *not closed*: every irrational number is a limit of rational numbers (e.g., $\sqrt{2} = \lim_n r_n$ for rationals r_n), but $\sqrt{2} \notin \mathbb{Q}$. So $\mathbb{Q}$ contains limit points not in $\mathbb{Q}$.

Q39. Which of the following is a correct statement of the Heine-Borel Theorem in $\mathbb{R}$?

- (A) Every bounded sequence has a convergent subsequence
- (B) A subset of $\mathbb{R}$ is compact iff it is closed and bounded
- (C) Every open cover of $\mathbb{R}$ has a finite subcover
- (D) Every continuous function on $\mathbb{R}$ is bounded

Correct Answer: (B)

The **Heine-Borel Theorem** (in $\mathbb{R}$ or $\mathbb{R}^n$): a subset K is *compact* if and only if K is *closed* and *bounded*. Equivalently, K is compact iff every open cover has a finite subcover. Note: (A) is Bolzano-Weierstrass, and (D) is the EVT (which requires compactness of the domain).

Quick Reference Summary Table

Theorem / Result	Statement (Key Points)
Completeness of $\mathbb{R}$	Every Cauchy sequence in $\mathbb{R}$ converges; every non-empty bounded set has a sup/inf.
Archimedean Property	$\forall x \in \mathbb{R}, \exists n \in \mathbb{N}$ with $n > x$.
Bolzano-Weierstrass	Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
Monotone Convergence	A monotone bounded sequence converges.
Cauchy Criterion	(a_n) converges $\iff (a_n)$ is Cauchy.
IVT	Continuous f on $[a, b]$, $f(a)$ and $f(b)$ opposite sign $\implies$ zero in (a, b) .
EVT	Continuous f on $[a, b]$ attains its max and min.
MVT	$f'(c) = \frac{f(b)-f(a)}{b-a}$ for some $c \in (a, b)$.
FTC	$\frac{d}{dx} \int_a^x f(t) dt = f(x)$ (when f continuous).
Uniform Convergence	Preserves continuity, allows interchange of lim and $\int$.
Weierstrass M-Test	$ f_n \leq M_n$ and $\sum M_n < \infty \implies \sum f_n$ uniformly convergent.
Heine-Borel	Compact in $\mathbb{R} \iff$ closed and bounded.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:

[Access the material here](#)

Good Luck with Your PSC Examination!

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